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Rebound tonometer having tilt correction

Abstract

A rebound tonometer is improved by providing a tilt signal from a tilt sensor of the rebound tonometer to a signal processor of the rebound tonometer, and configuring the signal processor to apply a tilt correction factor to a basic IOP measurement value calculated by the signal processor to provide a final IOP measurement value, wherein the tilt correction factor depends on a tilt direction and a degree of tilt indicated by the tilt signal. The final reported IOP measurement value takes into account gravitational effects on the probe due to tilt. As a result, measurements made at relatively small tilt angles are now usable, thereby improving efficiency. A corresponding rebound tonometry measurement method, and a method of calibrating a rebound tonometer for tilt correction, are also disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a division of U.S. application Ser. No. 16/645,523 filed Mar. 9, 2020, which is the U.S. national phase of International Application No. PCT/US2018/050346 filed Sep. 11, 2018, which claims priority benefit of U.S. Provisional Application No. 62/557,553 filed Sep. 12, 2017, the entire disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to rebound tonometers for measuring intraocular pressure (IOP).

BACKGROUND OF THE INVENTION

(2) A rebound tonometer is a hand-held instrument that propels a movable measurement probe in a controlled manner toward the cornea of an eye to measure intraocular pressure and/or corneal

biomechanics. The measurement probe is a disposable item typically having an elongated shaft terminating in a rounded tip. A new sterile measurement probe is loaded in the rebound tonometer prior to taking measurements on a patient. During a measurement, the probe contacts the cornea, is decelerated at a rate which depends on intraocular pressure, and then rebounds in a direction away from the cornea back toward the instrument housing. The rebound tonometer detects the motion of the measurement probe and determines intraocular pressure based on the detected motion of the probe. For example, the measurement probe may have a magnetized shaft which travels within a coil in the instrument housing. The coil may be energized momentarily to propel the probe toward the cornea by electromagnetic force, and then, after energizing current to the coil is shut off, a current may be induced in the coil by the moving probe to provide a detectable voltage signal (a measurement signal) representing motion of the probe. After measurements have been taken on a patient, the used measurement probe is discarded.

(3) The measurement accuracy of a rebound tonometer is dependent upon alignment of the instrument with the eye. Theoretically, for greatest accuracy, a travel axis of the probe (the measurement axis) should coincide with a central optical axis of the eye and the probe should travel a predetermined working distance along the measurement axis before contacting the eye at or very close to the corneal apex. To help with alignment and stability, it is known to provide an adjustable forehead support above the probe mechanism. The forehead support protrudes from the tonometer housing, and a distal end of the forehead support may be placed against the patient's forehead to establish a proper working distance. It is also known to equip a rebound tonometer with a sensing system capable of evaluating alignment and providing a yes or no indication of alignment to the user. Nevertheless, considerable skill and time is required to properly align the measurement axis to the eye. Because several (e.g. six) measurements may be recommended per eye and extra readings are often needed to refine alignment, rebound tonometry is sometimes considered inefficient.

(4) The position of the patient's head and direction of the patient's gaze may complicate alignment. If the patient's head is tilted and/or the patient's gaze is fixated along a direction that is inclined relative to horizontal, then the measurement axis cannot be properly aligned with the eye without tilting the rebound tonometer so the measurement axis is also inclined. However, if the measurement axis along which the probe travels is tilted to have a vertical component, the effects of gravity on the probe's motion may decrease measurement accuracy. U.S. Patent Application Publication No. 2016/0174838 teaches a rebound tonometer in which a tilt sensor (i.e. an inclinometer) is used to alert the operator prior to measurement that the measurement axis is inclined so that the operator may eliminate the inclination by repositioning the rebound tonometer and/or the patient so that the measurement axis is horizontal. However, if a measurement is made while the measurement axis is tilted, the accuracy of the measurement suffers due to the mentioned gravitational effects.

SUMMARY OF THE INVENTION

(5) A rebound tonometer is improved by providing a tilt signal from a tilt sensor of the rebound tonometer to a signal processor of the rebound tonometer, and configuring the signal processor to apply a tilt correction factor to a basic IOP measurement value calculated by the signal processor to provide a final IOP measurement value, wherein the tilt correction factor depends on a tilt direction and a degree of tilt indicated by the tilt signal. In this way, the final reported IOP measurement value takes into account gravitational effects on the probe due to tilt. As a result, measurements made at relatively small tilt angles are now usable, thereby improving efficiency and usability.

(6) A rebound tonometry method according to the present disclosure generally comprises the steps of operating a rebound tonometer to propel a measurement probe along a measurement axis toward an eye of a test subject such that the measurement probe is rebounded by the eye in a direction away from the eye, detecting measurement data describing motion of the measurement probe toward and away from the eye, sensing a direction and a degree of tilt of the measurement axis when the measurement probe is propelled toward the eye, calculating a basic IOP measurement

value from the measurement data, and applying a tilt correction factor to the basic IOP measurement value to yield a final IOP measurement value, wherein the tilt correction factor depends on the direction and the degree of tilt.

(7) The present disclosure also provides a method of calibrating a rebound tonometer generally comprising the steps of operating the rebound tonometer at predetermined tilt angles of the measurement axis to measure pressure of a simulated eye having a known pressure to determine a difference between the measured pressure and the known pressure, and storing information for determining an applicable tilt correction factor corresponding to each of the predetermined tilt angles, wherein application of the applicable tilt correction factor to the measured pressure at the corresponding predetermined tilt angle yields the known pressure, and wherein the stored information is available during normal use of the calibrated rebound tonometer.

Description

BRIEF DESCRIPTION OF THE DRAWING VIEWS

(1) The nature and mode of operation of the present invention will now be more fully described in the following detailed description of the invention taken with the accompanying drawing figures, in which:

(2) FIG. 1 is a schematic view of a rebound tonometer having tilt correction in accordance with an embodiment of the present invention

(3) FIG. 2 is a side view illustrating proper alignment of a rebound tonometer with an eye of a test subject, wherein a measurement axis of the rebound tonometer is horizontal;

(4) FIG. 3 is another side view also illustrating proper alignment of a rebound tonometer with an eye of a test subject, but the measurement axis is tilted from horizontal; and

(5) FIG. 4 is a flow diagram illustrating logic used by the rebound tonometer of FIG. 11 to compensate for effects of gravity due to tilting.

DETAILED DESCRIPTION OF THE INVENTION

(6) The present disclosure relates to improving a rebound tonometer by incorporating a tilt sensor to compensate for gravity effects when a measurement axis of a rebound tonometer is tilted from horizontal during a measurement.

(7) FIG. 1 is a schematic view showing a rebound tonometer **10** formed in accordance with an embodiment of the present invention. Rebound tonometer **10** generally comprises a disposable probe **12** and a hand-held housing **14** containing a measurement system **16** configured to propel probe **12** in a forward direction along a measurement axis **11** toward an eye of test subject, wherein probe **12** contacts a cornea C of the eye and is rebounded from the cornea in a reverse direction opposite the forward direction.

(8) Probe **12** may include an elongated shaft **12A**, at least a portion of which is made of a magnetic material, and a rounded tip **12B** at an end of shaft **12A** for contacting cornea C. Measurement system **16** may include a conductive drive coil **18** in which probe **12** is received, and a controller **20** configured to momentarily energize drive coil **18** to propel probe **12** forward toward the eye by electromagnetic force. Measurement system **16** may include a conductive measurement coil **22** through which probe **12** moves, and controller **20** may be further configured to measure a current induced in measurement coil **22** by the moving probe **12** and provide a measurement signal representing velocity of the probe as a function of time. For example, controller **20** may be configured to receive the current induced in measurement coil **22** by the moving probe **12** and provide an analog voltage signal as the measurement signal. The embodiment depicted in FIG. 1 shows drive coil **18** and measurement coil **22** as being two different conductive coils. Alternatively, a single coil may act sequentially during a measurement cycle as both the drive coil and the measurement coil, thus eliminating the need for a second coil.

(9) As known in the art of rebound tonometers, instrument **10** may further comprise an opto-electronic alignment detection system (not shown) and a display (not shown) to guide and confirm alignment of a measurement axis **11** of instrument with cornea C and positioning of a front nose **28** of instrument **10** at a predetermined working distance from cornea C. A trigger button **26** may be provided on housing **14** for enabling a user to send a signal to controller **20** to initiate a measurement, and/or the alignment detection system may automatically send a signal to controller **20** to initiate a measurement when alignment and proper working distance are confirmed by the alignment detection system.

(10) Measurement system **16** further includes a signal processor **24**, which may be part of controller **20** as shown in FIG. 1. Signal processor **24** may be configured to convert the analog measurement signal to digital form, and to calculate a basic IOP measurement value from the digitized measurement signal. For example, signal processor logic **20** may comprise an analog-to-digital signal converter and a programmed microprocessor for executing instructions stored in memory for calculating the basic IOP measurement value.

(11) Reference is now made to FIGS. 2 and 3. FIG. 2 is a side view illustrating proper alignment of rebound tonometer **10** with an eye of a test subject, wherein measurement axis **11** of rebound tonometer **10** is horizontal. As may be seen, gravity force “g” acts perpendicular to the direction of motion of the tonometer's rebound probe **12** along measurement axis **11**. Consequently, gravity force “g” does not have any component acting in the direction of motion of probe **12** that may influence the velocity at which the probe impacts the cornea and thereby affect the measurement result.

(12) FIG. 3 is another side view which also illustrates proper alignment of a rebound tonometer **10** with an eye of a test subject. However, in FIG. 3, the test subject's gaze direction is tilted slightly upward such that proper alignment requires that measurement axis **11** also be tilted from horizontal. In this situation, gravity force “g” includes a component acting in the direction of motion of the probe. As may be understood from FIG. 3, the gravity component will accelerate the probe as it travels toward the eye, causing the probe to impact the eye at a velocity greater than a desired predetermined velocity. Of course, if the test subject's gaze direction is tilted downward rather than upward, the probe will be decelerated by gravity as it travels toward the eye.

(13) In order to compensate for effects of gravity where measurement axis **11** is tilted from horizontal, rebound tonometer **10** may be equipped with a tilt sensor **40** as shown in FIG. 1. Tilt sensor **40** may be integrated in controller **20** or may be separate from controller **20** as depicted in FIG. 1. Tilt sensor **40** generates a tilt signal indicating a direction (i.e. upward or downward) and degree of tilt of measurement axis **11** at the time a measurement is initiated. Tilt sensor **40** may be connected to signal processor **24** for processing the measurement signal representing motion of probe **12** to determine IOP. The tilt signal from tilt sensor **40** may be provided to signal processor **24** and taken into account in the calculation of IOP to compensate for the unwanted effects of gravity on the measurement. By way of non-limiting example, tilt sensor **40** may be embodied as a Bosch Sensortec BMA253 triaxial, low-g acceleration sensor with digital output.

(14) FIG. 4 is a flow diagram illustrating an example of logic followed by signal processor **24** of rebound tonometer **10** to compensate for effects of gravity due to tilting. The logic may be embodied by software code stored in memory in rebound tonometer **10** and executed by signal processor **24**. As shown in block **110**, tilt sensor is monitored during an IOP measurement and the tilt angle of measurement axis **11** and probe **12** at the time of measurement is stored in memory. The IOP measurement is also taken and stored in memory pursuant to block **112**. Flow may then branch to one of three flow paths depending upon the sensed tilt angle. If the tilt angle is zero or nearly zero in accordance with block **114**, then the measured IOP is calculated without applying a tilt correction factor and is displayed as indicated by block **116**. If the tilt angle is between zero degrees and relatively low tilt angle, for example ± 30 degrees in accordance with block **118**, then the IOP is calculated and adjusted by applying a tilt correction factor based on the sensed tilt angle

to provide a final IOP measurement result, which is displayed as indicated by block **120**. The tilt correction factor may be empirically determined in a calibration step by operating tonometer **10** at predetermined tilt angles using a simulated eye of known pressure, and a look-up table of applicable correction factors for various tilt angles may be stored in memory for use in block **120**. Instead of a look-up table, a tilt correction function may be stored in memory for enabling calculation of an appropriate tilt correction factor based on the tilt signal. Application of the tilt correction factor may simply be addition of the tilt correction factor, which may be positive or negative, to the basic IOP measurement value. If the tilt angle is outside the acceptable range, for example its absolute value is greater than 30 degrees in accordance with block **122**, then an error message is displayed such as “Excessive Tilt—Retry” as indicated in block **124**.

(15) As will be appreciated from the foregoing description, rebound tonometer disclosed herein corrects for tilting of the measurement axis **11** during a measurement. This improvement allows more test subjects to be measured in a given period of time with greater measurement accuracy because slight inclination of the test subject's gaze does not disqualify measurements as in the prior art.

Claims

1. A method of calibrating a rebound tonometer operable to measure pressure of an eye by propelling a probe along a measurement axis toward the eye such that the probe contacts and rebounds from the eye, the method comprising the steps of: operating the rebound tonometer at predetermined tilt angles of the measurement axis to measure pressure of a simulated eye having a known pressure to determine a difference between the measured pressure and the known pressure; and storing information for determining an applicable tilt correction factor corresponding to each of the predetermined tilt angles, wherein application of the applicable tilt correction factor to the measured pressure at the corresponding predetermined tilt angle yields the known pressure, and wherein the stored information is available during normal use of the calibrated rebound tonometer.
 2. The method according to claim 1, wherein the stored information includes a look-up table of applicable tilt correction factors corresponding to the predetermined tilt angles.
 3. The method according to claim 1, wherein the stored information includes a tilt correction function for enabling calculation of an appropriate tilt correction factor based on the tilt signal.
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